

How to handle LLMs in your research

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Department of Methodology, Statistics & Data Science

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Today's Agenda

- Introduction (3 min)



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- How LLMs Work (10 min)



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- Limitations and Risks (10 min)
- Prompting CAs and Chatbots (10 min)
- Practical LLM Use in Research (10 min)



About us

Natural Language and Text Processing Lab



<https://nlp.sites.uu.nl/>

Introduction: What do you know about LLMs?

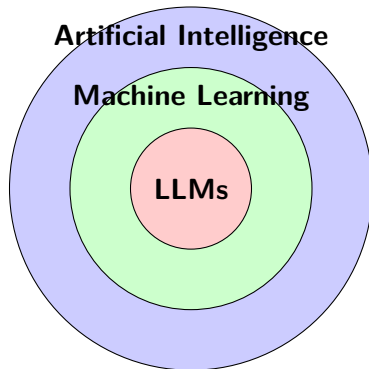
- **Activity: Think-Pair-Share** (2 min)



Image produced in DALL-E December 2023 (Credit: Christine Fox)



AI vs ML vs LLM: The Relationship



LLMs are a specific type of ML,
which is a subset of AI



Model Learning Process: Training Phase

Training Phase

- Model learns patterns from large datasets

Example: LLM learns that "cat" often appears with "dog" in pet-related contexts



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Model Learning Process: Training Phase

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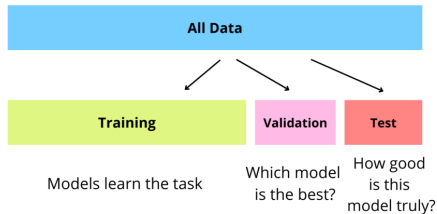
- Model learns patterns from large datasets
- Adjusts model to minimize prediction errors
- Requires massive computational resources (weeks / months)

Example: LLM learns that "cat" often appears with "dog" in pet-related contexts



Model Learning Process: Validation Phase

Validation and Testing



Source: <https://dqqlab.id/5-panduan-praktis-membuat-model-machine-learning>

Key insight: Good models **generalize**, not just **memorize**



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Prompting phase

U weet



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Credit for this slide: Albert Gatt & Dong Nguyen

Prompting phase

U weet wat



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Prompting phase

U weet wat ik



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What is Generative AI?

Definition: AI that creates new content (text, images, code, audio)

Key Features:

- Creates novel outputs, not just classifies

Examples: ChatGPT (text), DALL-E (images), CodeT5 (code)



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What is Generative AI?

Definition: AI that creates new content (text, images, code, audio)

Key Features:

- Creates novel outputs, not just classifies
- Learns patterns from training data
- Generates human-like content

Examples: ChatGPT (text), DALL-E (images), CodeT5 (code)



Text-GenAI components

	consists of	fed by	GPT learns
	LLM	textual internet data	“knowledge” of language and content
	Dialogue Model	social media, forums	“knowledge” of dialogue structure
	Task Model	programmers	tasks, self knowledge and limits
	Ethical Governor	Kenian data labellers	ethical answers

Credit for this slide: Davitze Könning



Generative AI vs. Other Models

Feature	Generative	Discriminative
Primary Task	Create new content	Classify existing data
Output Type	Novel text/images	Labels/categories
Example	GPT, DALL-E	Spam filters, medical diagnosis



Activity: True or False?

Instructions: Raise your hand if you think the statement is TRUE

- **Statement 1:** Generative AI can only work with text



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Caveat

gold labels!



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- **Statement 3:** Generative AI creates completely original content never seen in training



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- **Statement 3:** Generative AI creates completely original content never seen in training
- **Statement 4:** LLMs are a type of generative AI



Limitations and Risks: Hallucinations

What are hallucinations?

- LLMs generate false or nonsensical information (fake citations, non-existent facts)



Limitations and Risks: Hallucinations

What are hallucinations?

- LLMs generate false or nonsensical information (fake citations, non-existent facts)
- Appears confident and plausible



Bias

- **Gender bias:** Assumes doctors are male, nurses are female

Real-world impact: Biased models can perpetuate harmful stereotypes



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- **Cultural bias:** Primarily represents Western perspectives
- **Academic bias:** Favors published, English-language sources
- **Confirmation bias:** Reinforces existing beliefs

Real-world impact: Biased models can perpetuate harmful stereotypes



Limitations and Risks: Output Reliability

Reliability Issues

- **Inconsistency:** Same prompt can yield different answers

Give me the answer to the question "What is life?"

The question "What is life?" is a profound and complex one that has been explored by philosophers, scientists, theologians, and thinkers throughout history. There isn't a single, universally accepted answer, but here are some perspectives:

Give me the answer to the question "What is life?"

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Limitations and Risks: Output Reliability

Reliability Issues

- **Inconsistency:** Same prompt can yield different answers
- **Context dependency:** Performance varies by domain

Pre-Chorus:

Tus pasos suenan como un susurro,
En la noche, bajo el manto oscuro,
Y en tus brazos, siento el fuego puro,
De un amor que nunca se apuro.

Best practice: Always verify critical information from multiple sources

Limitations and Risks: Output Reliability

Reliability Issues

- **Inconsistency:** Same prompt can yield different answers
- **Context dependency:** Performance varies by domain
- **Prompt sensitivity:** Small changes affect outputs significantly

Write a Python program to do something basic



Certainly! Here is a simple Python program that calculates the sum of two numbers entered by the user:

This is a simple Python program to calculate the sum of two numbers

```
def main():  
    # Prompt the user to enter the first number  
    num1 = float(input("Enter the first number: "))  
  
    # Prompt the user to enter the second number  
    num2 = float(input("Enter the second number: "))  
  
    # Calculate the sum of the two numbers  
    sum_result = num1 + num2  
  
    # Print the result  
    print(f"The sum of {num1} and {num2} is {sum_result}")  
  
if __name__ == "__main__":  
    main()
```

Write a Python program to do something very basic



Certainly! Here is a simple Python program that prints "Hello, World!" to the console. This is a classic example often used to introduce programming beginners to the basics of a programming language:

This is a simple Python program to print "Hello, World!"

```
def main():  
    # Print "Hello, World!" to the console  
    print("Hello, World!")  
  
if __name__ == "__main__":  
    main()
```

Best practice: Always verify critical information from multiple sources

Limitations and Risks: Ethics

Ethical Concerns

- **Data:** Training data may include personal or copyrighted material

Limitations and Risks: Ethics

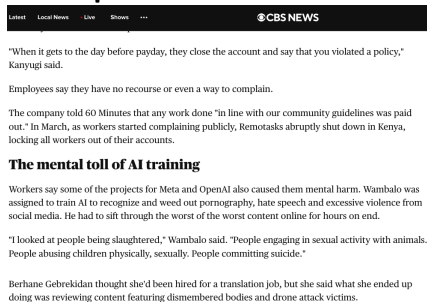
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<https://www.cbsnews.com/news/ai-work-kenya-exploitation-60-minutes> (Nov 2024)

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- **Open source:** Often little documentation on data, training, architecture
“Given both the competitive landscape and the safety implications of large-scale models like GPT-4, this report contains no further details about the architecture (including model size), hardware, training compute, dataset construction, training method, or similar”
OpenAI: Achiam, J. et al (2024). *GPT-4 Technical Report*.
<https://arxiv.org/abs/2303.08774>

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Results

Cultural modulation of neural mechanisms asserts that technological interactions significantly influence brain development and cognitive abilities. The use of artificial intelligence chatbots like ChatGPT for cognitive offloading may lead to underemployment of specific cognitive faculties, inhibiting their full maturation. This phenomenon is particularly relevant in the context of executive functions, where reliance on artificial intelligence for problem-solving can reduce cognitive effort and lead to long-term cognitive changes.¹³

- **Effect on cognition:**

Dubey et al. Redefining Cognitive Domains in the Era of ChatGPT: A Comprehensive Analysis of Artificial Intelligence's Influence and Future Implications. Med Res Arch. 2024 Jun;12(6):5383

Effective Prompting: BRAVE(R) Framework

Effective Prompting: BRAVE(R) Framework



B: Boundaries

Set limits on the format, length, or any other constraints.



R: Role

Identify the role or perspective you want the AI tool to take.



A: Audience

Specify who the output is intended for to determine the appropriate tone and style.



V: Variables

Highlight key details, variables, or points that should be included in the response.



E: Expectations

Clearly state what you expect the AI to accomplish, including the intended outcome and purpose.

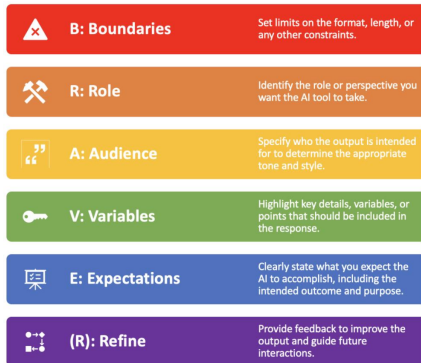


(R): Refine

Provide feedback to improve the output and guide future interactions.

Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).

Effective Prompting: BRAVE(R) Framework



Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).

Poor prompt: “Summarize this paper”

Better prompt: “Act as a psychology research assistant [R]. Summarize this paper in 3 bullet points [B] focusing on methodology and findings relevant to cognitive behavioral therapy [V]. Use academic language [A] and APA style [E].”

Key insight: Specific, contextual prompts yield better results

Evaluating and Critically Assessing GenAI Responses

Prompt formulation by Christine Fox, in collaboration with ChatGPT (May 2024).



F: Focus

Focus on the AI's response by breaking it down into its main points and arguments.



A: Authenticate

Authenticate the AI's output by verifying it against current research and empirical data.



C: Critique

Critique the response's accuracy, relevance, and depth in the context of the topic.



T: Think

Think about the response from various perspectives and its potential impact.



S: Scrutinise

Scrutinise the AI's conclusions by critically questioning and exploring alternative hypotheses or interpretations.

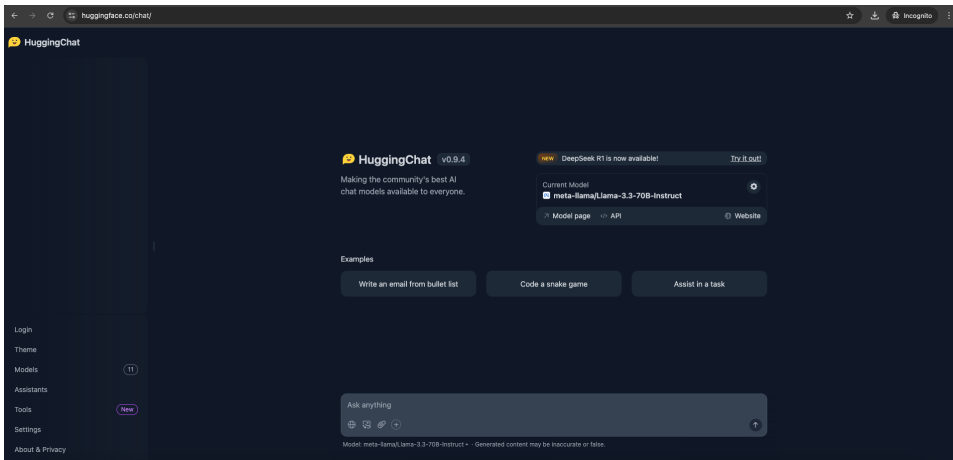
More info:

<https://teachersguides1.nl/genai/generative-ai-guidelines/>



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Ways to prompt



Switch models

×

Models

meta-llama/Llama-3.3-70B-Instruct

Active

Qwen/Qwen2.5-72B-Instruct

CohereForAI/c4ai-command-r-plus-08-2024

deepseek-ai/DeepSeek-R1-Distill-Qwen-32B

nvidia/Llama-3.1-Nemotron-70B-Instruct-HF

Qwen/QwQ-32B-Preview

Qwen/Qwen2.5-Coder-32B-Instruct

meta-llama/Llama-3.2-11B-Vision-Instruct

NousResearch/Hermes-3-Llama-3.1-8B

mistralai/Mistral-Nemo-Instruct-2407

microsoft/Phi-3.5-mini-instruct

Assistants

My Assistants

meta-llama/Llama-3.3-70B-Instruct

Ideal for everyday use. A fast and extremely capable model matching closed source models' capabilities. Now with the latest Llama 3.3 weights!

➤ Model page ➤ Model website </> API Playground 🔗 Copy direct link to model

💬 New chat

System Prompt

Official UU policy

<https://www.uu.nl/en/organisation/ai-policy/researchers>

- Benefits and use cases



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- Benefits and use cases
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- Effective use of GenAI (*Background check*)



Official UU policy

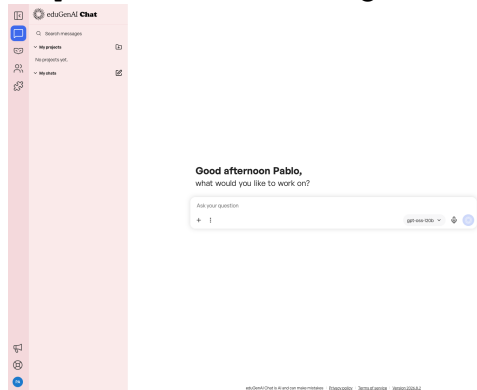
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- Benefits and use cases
- Risks and limitations
- Effective use of GenAI (*Background check*)
- Citing GenAI in your work



EduGenAI

<https://www.uu.nl/en/organisation/ai-policy/teachers/edugenai>



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UU AI-chat

<https://www.uu.nl/en/organisation/ai-policy/ai-platforms-at-uu>

UU AI-chat

UU AI-Chat is Utrecht University's own generative AI chat environment. You use it via your browser to have conversations with various language models, within an environment managed by the University itself. The platform is based on the University of Amsterdam's AI chat environment. UU AI-Chat will be available in early September.

Who is it for?

It is intended for all students and staff at Utrecht University, for use in teaching, research and administrative operations. Guests and external parties cannot use the service.

What can you do with it?



More about UU AI Chat



How do you gain access?

This service will be available to students and staff from the start of September.



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LLM Use in Research

Appropriate & Inappropriate Uses

- Think-Pair-Share



LLM Use in Research: Appropriate Applications

Appropriate Uses

- **Literature review:** Summarizing papers (with verification)

Key: Always maintain human oversight and verification



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Caveat

only when you know which statistical analysis is suitable

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Caveat

only when you know which statistical analysis is suitable

- **Brainstorming:** Generating research ideas
- **Translation:** Working with multilingual sources

Key: Always maintain human oversight and verification



LLM Use in Research: Inappropriate Applications

Avoid Using LLMs For:

- **Statistical analysis:** LLMs can't perform proper statistical tests



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- **Statistical analysis:** LLMs can't perform proper statistical tests
- **Fact-checking:** They may hallucinate information
- **Critical decisions:** Participant safety, ethical approvals
- **Original thinking:** Generating truly novel theories
- **Replacing expertise:** They supplement but don't replace researcher knowledge



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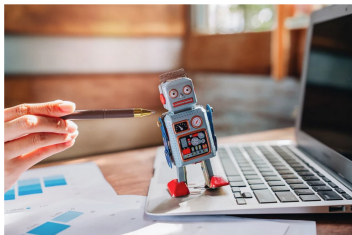
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By Yanjun Shen



Using generative artificial intelligence models to kick ideas around is more effective than outsourcing thinking wholesale, says Yanjun Shen. Credit: iahkittirha/Stock via Getty

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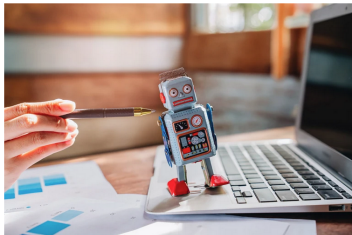
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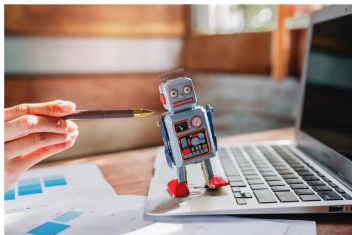
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① thinking for yourself

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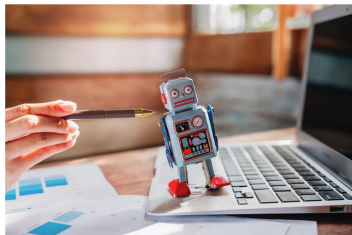
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- ① thinking for yourself
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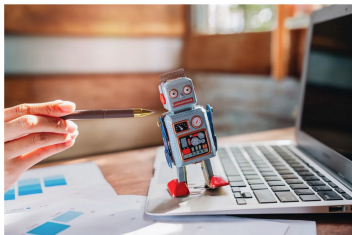
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- 2 read literature
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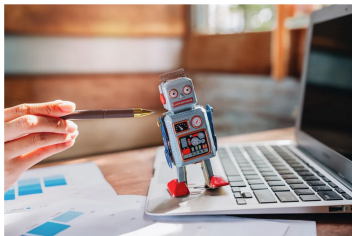
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- ① thinking for yourself
- ② read literature
- ③ ask AI for advice
- ④ critically analyze